Category Poster

Title (U) Provable Security: A Hybrid Machine Learning and Formal Methods Approach for Software Verification

Presenter William Bardon, UMBC, TS/SCI

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Most applicable topic areas Research and Development - Automated Software Analysis Tools

(U) Provable Security

A Hybrid Machine Learning and Formal Methods Approach for Software Verification

William Bardon b277@umbc.edu
Charles Nicholas nicholas@umbc.edu
Department of Computer Science and Electrical Engineering
University of Maryland, Baltimore County (UMBC)
Baltimore, Maryland 21250 USA

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(U) Abstract

(U) Generative AI is already producing code faster than humans can review and fix. Formal Methods (FM) can verify algorithms and protocols, but they must be encoded into model syntax. Mastering FM model syntax is its own challenge. FM tools are definitive if they reach completion, but they may run out of memory, storage, or time before then. Understandably, Formal Methods have not been widely adopted outside of safety-critical applications such as healthcare or spaceflight.

(U) The strengths of Generative AI and Large Language Models (LLMs) complement weaknesses of Formal Methods. AI models require immense computing resources and huge amounts of data to train, but once trained, they produce responses relatively quickly. AI always produces a result, and LLMs respond to prompts in ordinary language. Unfortunately, LLMs hallucinate or create invalid constructs that look similar to real programs. They mimic training data and their memory and attention spans are limited.

(U) We propose a hybrid multi-step solution. AI can translate prompts into code, so it can also translate requirements or specifications into FM models for verification. FM tools then verify models and provide feedback to AI to revise models if needed. Finally, AI can translate FM models back into specifications or code. Formal Methods and Generative AI can each compensate for weaknesses in the other.